

Digital Justice

As the lines between real and virtual worlds blur, our *avatars* need to be as alert as us, because the dangers of the real world are bound to occur in the virtual one too. We need to be aware of cyber frauds and privacy infringement. Cases of groping on MR platforms have also been reported. Soon after Meta launched Horizon Worlds, a beta tester reported on their website that she had been groped by a stranger on Horizon Worlds. Meta reviewed the incident internally, and suggested that this could have been averted if the tester had used a tool called Safe Zone, which is a protective bubble users can activate when feeling threatened. Within it, no one can touch them, talk to them, or interact in any way until they signal that they would like the Safe Zone lifted. While some platforms have such features in place, many do not. And even where such features are available, most users do not know about it. So, more awareness has to be created.

Tech futurist and author Dr Ayesha Khanna suggested in one of her interviews that all virtual worlds must have some systems in place for digital justice and good governance, to prevent digital harm and cyber-crime, and to punish such occurrences if they occur despite best efforts.

cities and locations, where players can chill out in 3D versions of their favourite hangouts like Connaught Place in Delhi, Marine Drive in Mumbai, or MG Road in Bengaluru.

There are also virtual spaces for more serious work, like Microsoft Mesh and Meta's Horizon Workplaces for collaborative meetings and design sessions, or NextMeet for meetups, and VRXtream for organising events.

Meta hopes to also bring you closer to your family and friends with Horizon Worlds and Horizon Homes, where, rather than fighting to save the world, you will be building escape rooms, social hangout spaces and party games to foster relationships.

So, now we know that there are multiple virtual worlds in this metaverse, each with different themes and purposes. If you look at it very objectively, these worlds are nothing but interoperable software platforms that you log in to for work or play. What makes them worthy of being called 'worlds' is their immersive nature, and ability to host thousands of people in a single instance of the server.

These worlds are immersive, and you enter them as *avatars*

Take Zoom or Google Meet—these are also ways of interacting with people. But everyone, including you, is just a thumbnail in it. You can see and talk to each other, but you are not really in the same space. The

metaverse, on the other hand, is immersive and experiential. You enter this 3D digital space and become part of it, along with others. So, if you are playing with friends in Horizon Worlds, your *avatars* will be together in the same park or beach. If you are attending a math session, you will be sitting in a virtual classroom next to your friend.

So, to experience the metaverse, you first need an *avatar*, and then you need the tech gear to immerse yourself in the virtual world. Your *avatar* is, simply put, a 3D character that represents you in the digital world. You can create your own *avatar*, or customise available ones, most of which look like animated characters. However, the more serious and real-world applications emerging today will feel better if your *avatar* looks something like you. So, Epic Games owned company Unreal Engine has developed a tool called MetaHuman Creator, which would help you develop digital humans that look and behave pretty much like you.

Once you have an *avatar*, you enter the virtual world. "Metaverse can be experienced using personal computers (PCs), laptops, mobiles, etc. Virtual reality (VR) and augmented reality (AR) are hardware technologies used to experience the metaverse. With VR, you are in a 360-degree immersive environment, whereas with AR you augment the 3D elements with the real world," explains Kypuram. Going a step

forward, we now have mixed reality (MR)—a form of AR where the real world and digital objects are not just visually interleaved but also interact with each other.

As the Rs (VR, AR, MR) keep increasing, we see them being brought under one umbrella term called extended reality (XR). According to VRdirect, the term simply represents any sort of technology that alters reality for the user by blurring the line between real and digital worlds. So, it combines VR, AR, and MR, as well as any new technologies that will come up in the future.

VR/MR headsets need no introduction to gaming enthusiasts—Oculus from Meta, HoloLens from Microsoft, and Playstation VR headsets from Sony, to name a few. VR headsets, most commonly used by serious gamers, will totally immerse you in a simulated environment using 3D vision, spatial sound, and motion trackers that sense movement of your eyeballs, and sometimes even jaw movements. The headsets usually come with handheld gaming controllers too. Some advanced sets might also include cameras to detect full body movement, and haptic feedback vests to give you a sense of touch—so you can feel it when your opponent punches you!

MR headsets go a step further to combine real-world and digital elements, usually using holographic display tech. Not an easy task, and right now they cost a whopping lot, but MR headsets might be the actual solution to balancing real and digital lives. Betting big on the metaverse, most of the industry majors are working on developing advanced MR headsets. Last year, Meta revealed two such projects—Project Nazare and Project Cambria. Apple is also developing an MR headset. Microsoft HoloLens 2 is a preferred MR headset for professional and industrial uses—but its lowest-end model costs most than US\$3,000.

The problem with most headsets